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ORIGINAL PAPER

The Fork Game: A Graphical Interface for Eliciting Higher-Order Risk Preferences

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Abstract

We introduce the "Fork Game," a graphical interface designed to elicit higher-order risk preferences. In this game, participants connect forked pipes to create a final structure. A ball is then dropped into the top opening of this structure and follows a downward path, randomly turning left or right at each forked joint. This construction is effectively isomorphic to the apportionment of binary-outcome lotteries, allowing participants to construct complex gambles. Furthermore, the game is easily comprehensible, highly modular, and provides a flexible means of assessing risk aversion, prudence, temperance, and even higher-order risk preferences.

Keywords: Experiment; Prudence and temperance; higher order risk preferences

JEL Codes: C91; C93; D14; D81; E21

1. Introduction

A major research program in economics focuses on the development of incentivized laboratory tasks to elicit individuals' attitudes towards risk. It has been generally shown that most individuals can be characterized as risk averse in the gain domain. Risk-averse decision makers are willing to forgo some amount of their initial wealth to avoid certain actuarially fair risks, such as mean-preserving spreads (Rothschild and Stiglitz, 1978). When preferences over risky prospects admit the classical expected utility framework and are represented by an increasing and continuously differentiable von Neumann-Morgenstern utility function, risk aversion can be characterized by the second derivative of this function being negative. The signs of the higher-order derivatives of the utility function have further implications for the behavior of decision makers under uncertainty. Take the case of precautionary savings, realized when an increase in risks to future income is offset by an increase in present day savings. Leland (1968) and Sandmo (1970) show that when the third derivative of this utility function is positive, that is when the marginal utility is convex, a risk-averse individual will increase her demand of precautionary savings.

Kimball (1990) coined the term *prudence* for this type of behavior. In Kimball (1992), a related concept of *temperance* is defined as the willingness to refrain from further risks to income, when the agent is already facing an unavoidable risk. He also mentions that this is similar to the *properness* concept of Pratt and Zeckhauser (1987), which describes an agent facing two undesirable lotteries, to continue classifying one of these as undesirable when he is forced to face the other one.

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In their seminal work, Eeckhoudt and Schlesinger (2006) discovered that prudence, temperance, as well as any further risk apportionment of order n, can be characterized by preferences over simple lottery pairs. Furthermore, they showed that these preferences are equivalent to the sign of the higher-order derivatives of the von Neumann-Morgenstern utility function. The nature and simplicity of these lottery pairs have motivated many further experimental studies aiming to assess higher-order risk preferences and their relation to real-life risk-taking behaviors. Their framework also allows for a parameter-free comparison of higher-order risk preferences across individuals by constructing indexes based on the number of certain binary decisions (Trautmann and van de Kuilen, 2018).

While early studies focused on identifying the sign of higher-order derivatives of the utility function through binary choices between simple lottery pairs, more recent work propose methods to measure the degree or intensity of these preferences. Schneider and Sutter (2020) propose a methodology that uses certainty equivalents to construct utility functions non-parametrically, allowing for measurement of the strength of higher-order risk attitudes. Haering et al. (2020) and Haering (2021) examine the consistency of higher-order risk preferences across different presentation formats, highlighting the importance of experimental design in preference elicitation. These methodological advances underscore both the significance of measuring higher-order risk preferences and the challenges in their elicitation.

In this paper, we introduce the "*Fork Game*," a novel method that utilizes forking pipes to represent the randomization aspect of the experiments. The resulting outcome is represented by a falling ball within those pipes, taking a randomly chosen path at the forks. In the treatments, participants are tasked with placing the pipes to construct the desired lottery, which is equivalent to the lottery pairs used in previous literature. By visualizing the randomization aspect and the chosen lottery clearly, and by designing all games as direct variations of the same underlying concepts, we can extend the games in a straightforward manner to analyze even more complex risk attitudes without significantly increasing the procedural complexity.

Tasks designed to elicit higher-order risk preferences naturally involve complex gambles. In these environments, research has documented that subjects exhibit significantly divergent decision-making patterns when confronted with complex gambles compared to their reduced counterparts, highlighting the challenge of comprehending the overall consequences of such choices (Haering et al., 2020). Another factor to consider is task difficulty. We believe our task is relatively simple, as approximately 8% of participants rated its difficulty as 0 on a 0-10 scale, and no subject rated it above 8. The average reported difficulty was 3.7, with a median of 3. In addition to the Fork Game, we also experimentally replicated Bleichrodt and Bruggen (2022) and Noussair et al. (2014) and compared the results of these tasks in three dimensions: observed choices for risk aversion, prudence, and temperance; distributional patterns of these choices; and operational aspects, including task difficulty and completion times. Our results indicate broadly consistent risk preferences across all designs, with participants making approximately 7-8 risk-averse choices out of 12, with no significant differences. Notable differences arise in prudence, where our Fork Game and Noussair et al. (2014) exhibit significantly higher prudent choices compared to Bleichrodt and Bruggen (2022). Temperate choices show consistency across all designs, with participants making temperate decisions in about 5-6 cases. The graphical similarity between the Fork Game and Noussair et al. (2014), along with the similarity of observed choices, indicates that design plays a significant role in elicitation. The Fork Game exhibited faster completion times, particularly for prudence-related tasks, and comparable or lower completion times for temperance, suggesting its potential to be a handy tool for tasks aimed at measuring higher-order risk preferences. This efficiency, combined with its simplicity, highlights the method's suitability for large-scale experiments where time constraints are critical.

While risk aversion remains central in the literature on economic decisions under uncertainty, higher-order risk preferences have been demonstrated to complement the complete characterization

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Fig. 1 Prudence according to Eeckhoudt and Schlesinger (2006)

of economic behavior across various domains. For instance, as stated previously, the precautionary savings motive, which entails increased savings in response to higher future income risks, is associated with the convexity of marginal utility, denoted as $u''' > 0$.[1] Evidence from Esö and White (2004) supports the existence of precautionary behavior in auctions, where the introduction of background risk leads prudent bidders to reduce their bids by a greater extent than the risk premium, termed as precautionary bidding. Treich (2010) establishes a theoretical relationship between risk aversion and rent-seeking behavior, demonstrating that risk aversion reduces rent-seeking only among prudent individuals in the context of rent-seeking games. White (2008) examines the impact of prudence in a bargaining game under risky outcomes. Notably, prudence emerges as a more significant determinant of health expenditures compared to risk aversion, as evidenced in studies by Courbage and Rey (2006), Krieger and Mayrhofer (2017). Within the financial literature, Brunnermeier et al. (2007) find that individuals, in line with prudent behavior, tend to overinvest in positively skewed assets. Schneider (2019) introduces a decomposition of forward market returns, revealing that downside risk contributes significantly to the forward premium, aligning with prudent behavior.

The remainder of the paper is organized as follows: Section 2 provides a technical characterization of risk aversion, prudence, and temperance. Section 3 describes the experimental tasks used in the literature to elicit higher-order risk preferences. Sections 4 and 5 outline the design aspects of the "Fork Game" and the experimental procedures, respectively, and Section 6 presents an empirical evaluation of the experiment's results. Finally, Section 8 concludes with a summary and discussion of the results.

2. Risk aversion, prudence and temperance

We start with a technical characterization of risk aversion, prudence and temperance using the framework in Eeckhoudt and Schlesinger (2006). Consider a decision maker whose preferences admit the classical expected utility representation, with u being the von Neumann-Morgenstern utility function (Von Neumann and Morgenstern, 1953). Assume further that u is an increasing and continuously differentiable function. Let w be the initial wealth of this decision maker. Let $k, \delta > 0$ be two constants and $\{\tilde{\epsilon}_i\}$ be a list of mutually independent and zero-mean random variables. The weak preference relation over lottery pairs is represented by $\succeq$, and we use $[o_1, o_2]$ represent a lottery with two possible and equally likely outcomes o_1 and o_2. The decision maker is risk averse if $w \succeq w + \tilde{\epsilon}_i$, $\forall w$ and $\forall \tilde{\epsilon}_i$. This is equivalent to the second derivative of u being negative. Risk aversion also implies that the individual prefers to face two certain losses at different states of the world, that is, $[w-k, w-\delta] \succeq [w, w-k-\delta]$, $\forall k, \delta > 0$.[2] The decision maker is said to be prudent if she prefers to disaggregate a sure loss and a zero-mean lottery, that is she prefers to face them in different states of the world. Formally, the preferences of a prudent decision maker imply $[w - k, w + \tilde{\epsilon}_i] \succeq [w, w - k + \tilde{\epsilon}_i]$, $\forall w$, k, and $\forall \tilde{\epsilon}_i$, as demonstrated in Figure 1. Eeckhoudt and Schlesinger (2006) show that this is equivalent to third

[1] Leland (1968), Sandmo (1970), Dreze and Modigliani (1972).

[2] As we are assuming a Von Neumann–Morgenstern utility representation, directly follows from Jensen's Inequality Jensen (1906).

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Fig. 2 Temperance according to Eeckhoudt and Schlesinger (2006)

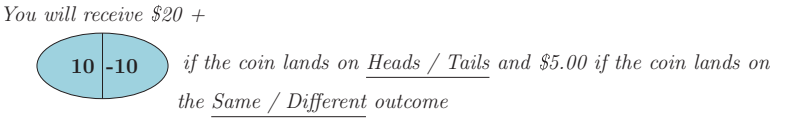


Fig. 3 A compound lottery in Deck and Schlesinger (2010)

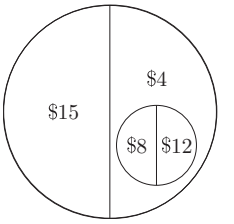


Fig. 4 A compound lottery in Deck and Schlesinger (2014)

derivative of u being non-negative, i.e. having a convex marginal utility function. In their words, adding $\bar{\epsilon}_i$ to a higher wealth is less painful for a prudent individual.

On the other hand, the decision is maker is said to be temperate if she prefers to face two statistically independent zero-mean lotteries, $\bar{\epsilon}_i$ and $\bar{\epsilon}_j$, at different states of the world. Formally, the preferences of a temperate decision-maker imply $[w + \bar{\epsilon}_i, w + \bar{\epsilon}_j] \succeq [w, w + \bar{\epsilon}_i + \bar{\epsilon}_j]$, $\forall w$, $\forall \bar{\epsilon}_i$, and $\forall \bar{\epsilon}_j$, as demonstrated in Figure 2. Eeckhoudt and Schlesinger (2006) show that this equivalent to fourth derivative of u being less than or equal to zero.

3. Experimental tasks from previous literature

We begin by outlining the design aspects of key experimental studies that examine higher-order risk preferences, followed by a summary of their behavioral outcomes.

Deck and Schlesinger (2010) use a text-based design supplemented with figures representing binary lotteries. Let $[+\epsilon, -\epsilon]$ be the zero mean lottery and $k > 0$ be a constant. In their prudence tasks, participants start with an initial amount ($w > 0$) and are presented with two equally likely states, Heads or Tails, determined by a coin flip. They are instructed to (i) choose the state under which the outcome of the zero mean lottery would be added to their initial amount, and (ii) select the state in which they would prefer an additional fixed amount (k) to be added to their earnings. For temperance tasks, a different zero mean lottery is used instead of k, and the choice procedure remains similar. The outcome of binary lotteries is determined by a spinner with a half-green (high payoff) and half-red (low payoff) configuration. Figure 3 provides an example task where participants circle their choices in the underlined sections of the italicized text. By circling "Head" (or "Tails") along with "Same" in this task, participants exhibit prudence as they are willing to face the zero mean lottery in the favorable state. On the other hand, subjects circling "Different" instead of "Same" exhibit imprudence in this task.

In Deck and Schlesinger (2014), compound lotteries are depicted using a representation similar to Figure 4 below. The divided portions of a pie shape correspond to different states, while a smaller

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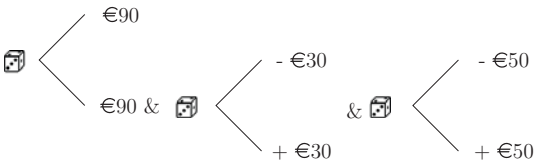


Fig. 5 A compound lottery in Noussair et al. (2014)

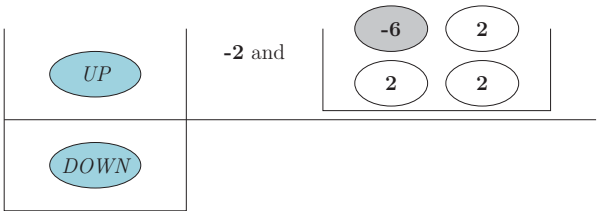


Fig. 6 A compound lottery in Ebert and Wiesen (2011)

pie within a larger one represents an additional lottery accompanying the realization of a specific state. The outcomes of these lotteries are determined using a spinner, similar to the approach in Deck and Schlesinger (2010). This design allows for the combination of lotteries by drawing smaller pie shapes inside larger ones, enabling the construction of complex gambles. In their study, participants are presented with a choice between two compound lotteries for each task, resulting in a total of 38 tasks.

In Noussair et al. (2014), compound lotteries are represented as shown in Figure 5 below. In this particular compound lottery, subjects first engage in a binary lottery with two equally likely outcomes. The first outcome (€90) occurs if the dice throw is equal to 1, 2, or 3, while the second outcome (also €90 in this example) occurs if the dice throw is equal to 4, 5, or 6. In the latter case, subjects face two additional binary lotteries, each with two equally likely outcomes determined by separate dice throws. This example illustrates a scenario where two zero mean lotteries are encountered in the same state, representing an intemperate choice. The alternative option involves facing these gambles in different states, with the second zero mean lottery, with respective outcomes €50 and -€50, received in tandem with the realization of the first outcome of the initial lottery (when the first dice throw is equal to 1, 2, or 3). This combination would indicate a temperate choice.

A simplified representation of the compound lotteries used in Ebert and Wiesen (2011) is shown in Figure 6 : the initial 50/50 gamble is depicted as a ballot box containing two balls labeled "Up" and "Down." If the "Up" ball is drawn, the subject incurs a loss of 2, and a second zero-mean risk lottery follows. The second lottery is represented by another ballot box containing multiple balls, with four balls shown in this simplified version. Yellow balls indicate a loss, while white balls indicate a gain. If the "Down" ball is drawn from the first ballot box, no loss occurs, and no second lottery follows. A preference for this particular compound lottery would indicate an imprudent choice.

In a recent experimental study, Bleichrodt and Bruggen (2022) investigate whether higher-order risk preferences exhibit differences between gain and loss domains, and examine the existence of a reflection effect similar to the one observed for risk aversion. Figure 7 presents a simplified representation of the options used in their experiment. In this representation, lotteries with binary outcomes are depicted as the act of drawing a random ball from an urn. In the given example, the subject is confronted with a binary lottery offering a payout of €5 or €10. If the latter outcome is realized, the subject then faces an additional binary lottery where

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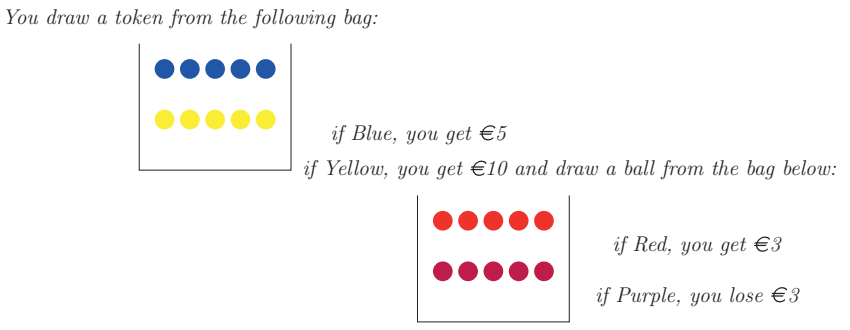


Fig. 7 A compound lottery in Bleichrodt and Bruggen (2022)

there is an equal probability of winning or losing €3. This combination would indicate a prudent choice since the subject faces the second lottery in tandem with the good outcome of the first lottery.

In addition to the lottery pairs approach described in the previous examples, alternative methods have been employed for the experimental measurement of higher-order risk preferences. In Ebert and Wiesen (2014), a multiple price list approach is utilized, where uncertainties are resolved through draws from ballot boxes. The graphical representations of these lotteries are similar to Figure 6, but each time a subject compares two such lotteries, varying amounts of compensation are added to one of the two lotteries, forming a price list. Another approach is presented in Schneider et al. (2022), where certainty equivalents are initially elicited and used to derive utility points. These utility points are then employed in a spline regression to construct a non-parametric utility function. From this function, n^{th} order derivatives can be obtained, enabling the classification of higher-order risk preferences. This methodology is also applied by Schneider and Sutter (2020).

The experimental findings from Deck and Schlesinger (2010) indicate evidence of prudence among subjects, although to a limited extent. However, intemperate behavior is observed more frequently than temperance. In their subsequent study, Deck and Schlesinger (2014) find that subjects display a strong inclination towards risk aversion, prudence, and a moderate preference for temperance. They also identify a significant correlation between risk-averse and temperate behavior. Haering et al. (2020) replicate this setup among subjects from the USA, China, and Germany and they observe a behavior similar to the findings of Deck and Schlesinger (2014). They also explore the reduced-form equivalents of compound lotteries for a subset of participants, highlighting certain behavioral patterns, such as reduced prevalence of prudent and temperate choices, that are influenced by this framing.

Noussair et al. (2014) reports similar findings in their sample, which includes both individuals from the general population and a group of students. They observe a high degree of prudence among both groups, while temperance is less prevalent. Ebert and Wiesen (2011) observe an aggregate level of prudence comparable to that reported in Deck and Schlesinger (2010). They also find that most prudent individuals exhibit a skewness seeking behavior, while the reverse may not generaly hold. Bleichrodt and Bruggen (2022) examine higher-order risk attitudes under three treatments: losses, 50-50 gains, and small probability gains. The 50-50 gains treatment aligns with previous studies utilizing binary lotteries. In this treatment, subjects exhibit significant risk aversion, and moderate prudence, but also a tendency toward intemperate behavior.

In summary, the general results from these studies suggest that prudence is commonly observed, while the presence of temperance is less consistent. Several studies demonstrate a higher frequency of intemperate behavior. A comprehensive review of the literature on higher-order risk preferences can be found in Trautmann and van de Kuilen (2018).

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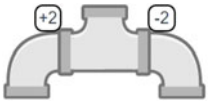


Fig. 8 A simple lottery, represented with a pipe

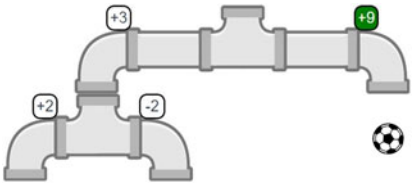


Fig. 9 End result for a prudence round

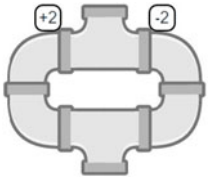


Fig. 10 Pipes for the temperance treatment

4. Fork Game: Design

In our experimental design, lotteries are represented by pipes, which were typically forked as in Figure 8, with some variations on their horizontal size. In this figure, a lottery with two equally likely outcomes that has the payments of $[2, -2]$ is represented.[3]

In all the games corresponding to each of the concepts, the resolution of uncertainty is represented by a falling ball which goes left or right with probability 50% whenever it encounters a fork on its path.

The task in each round of the Risk Aversion treatment is straightforward and the subject is asked to choose one forking pipe amongst two, differing in the payoffs shown in their tags. The main advantage of our design does not show itself in this game, but it serves as a nice introduction for the next two games, which will use the same image files and similar designs.

In a given round of the Prudence treatment, the subject is expected to place a small forking pipe after one of the two ends of a big forking pipe. The first and bigger pipe represents the first lottery with each side assigned a payoff as usual. The slots to place the smaller pipe are aligned directly under the exits of the bigger pipe, so the ball can continue its path, if it happens to reach the second pipe.

In Figure 9, a sample round for Prudence treatment is shown. The subject can construct one of two distinct lotteries by placing the smaller pipe to the slot at the left end or the right end of the larger pipe: $[(3 + [2, -2]), 9]$ and $[3, (9 + [2, -2])]$. In this example, we see that the subject went with the former. From the figure, we (and in the experiment the participant) also see the result of this specific round, since the ball reached to the right hand side, gaining the participant payoff of 9 for this specific round.

Finally, for our Temperance treatment we introduce another type of pipe, as shown in Figure 10. The main point of introducing this pipe is to allow sublotteries to be placed after one another, independent from their result. While the payoff in this case is same with the case with Figure 8, here there is additional utility that we can chain one lottery after another.

[3]We used a modified version of the image files that are available at https://www.kenney.nl/assets/puzzle-pack-2 with a Creative Commons CC0 license. Mentioned modifications are done with Inkscape, an open source tool.

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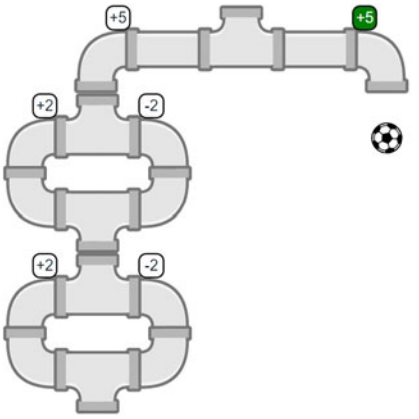


Fig. 11 End result for a temperance round

In the Temperance treatment, the participant has to choose the placement for two forking pipes given, and there are four slots to choose from. However not all subsets of these four slots are available for selection, since the below two slots are only available if the slot directly above is already filled.

In Figure 11, the end of an example temperance round is shown. Here, the participant is presented two lotteries:$[(5 + [2, -2]), (5 + [2, -2])]$ and $[5, (5 + [2, -2] + [2, -2])]$ and from what we observe they went with the latter one. Note that, the payoffs are symmetric here, so it does not make a difference if the lotteries are placed one after another in the right side or the left side. The random selection chose the right side, resulting in a total payoff of 5. If left side of this contraption was chosen randomly instead, they would face with two additional lotteries of $[2, -2]$.

This setup is easily extendable to represent risk apportionment of any degree so as to elicit even higher order risk preferences. In the Online Appendix, we describe both theoretically and graphically how the Fork Game can be used to elicit edginess attitudes, which is equivalent to 5th derivative of the respective von Neumann-Morgenstern utility function, $u(.)$, being positive.

5. Experimental procedures

The experiments were conducted in the Economics Laboratory at Boğaziçi University in two separate phases. The first phase took place between December 2022 and March 2023, while the second phase occurred between November 2024 and December 2024. The experimental sessions in the first phase involved the Fork Game, whereas those in the second phase included either the Fork Game, or a replication of the experiment in Bleichrodt and Bruggen (2022), or a replication of the experiment in Noussair et al. (2014). In replicating Bleichrodt and Bruggen (2022) and Noussair et al. (2014), we closely followed their original design aspects. However, in our implementation, uncertainties were resolved immediately, and subjects observed their round-specific earnings at the end of each round. This design choice ensured consistency across all experimental formats in our study. For example, during the replication of Bleichrodt and Bruggen (2022), the balls selected from the urns were highlighted to indicate the outcome.

Participants were recruited via the online recruiting system ORSEE (Greiner, 2015). A total of 402 subjects participated in the experiments: 134 in the first phase and 268 in the second phase, over 67 sessions.[4] All participants were students at Boğaziçi University, with an average age of 21.01. Among them, 47.3% identified as female and 52.7% as male.

[4]Fork Game participants from both phases show consistent choice patterns and comparable demographic characteristics. Random assignment in the second phase produced balanced treatment groups.

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In all experiments, participants first encountered the Risk Aversion treatment, followed by the Prudence and Temperance treatments. The order of these latter two treatments was randomized for each participant, as in Noussair et al. (2014). Lotteries within each treatment were also presented in a random order. Additionally, the graphical placement of options and the placement of potential rewards within these options were randomized. For instance, in the Fork Game, rewards were randomly assigned to the forking pipes and then for each lottery pair, the values were assigned to the left or right side of a pipe with equal probabilities.

Each treatment consisted of 12 rounds. The payoff structure was based on the 50–50 gains treatments described in Bleichrodt and Bruggen (2022), with the full set of parameters detailed in the Online Appendix. Payoffs were presented as "points" in all experiments. After completing the three treatments, participants answered demographic questions and completed the full set of the Global Preferences Survey. All treatments and survey questions were presented in Turkish.

The experimental software was developed as a web application in JavaScript using the Vue.js framework. It was deployed on GitHub and accessed via standard Chrome browsers in full-screen mode.[5] The software also included the main introduction and separate instructions for each of the three treatments.

Participant compensation included earnings from one randomly selected round plus a participation fee. To account for Turkey's significant inflation, we adjusted the point-to-currency conversion: in the first phase, each point equaled 1.5 Turkish Lira (TL) with a 20 TL participation fee, while the second phase used 3 TL per point with a 40 TL participation fee. Average earnings including participation fee were 52.5 TL for the first phase, and 105.6 TL for the second phase. The Global Preferences Survey's monetary values, originally from the 2012 Gallup World Poll, were also inflation-adjusted: multiplied by 5 for the first phase and by 10 for the second phase.[6]

6. Results

We begin by presenting the incidence of risk-averse, prudent, and temperate choices in the Fork Game experiment. Table 1 displays the average number of risk-averse, prudent, and temperate choices based on the count of binary choices made out of the 12 tasks in each treatment. To assess the prevalence of each risk attitude, we adopt the approach used in Deck and Schlesinger (2010), Ebert and Wiesen (2011), and Noussair et al. (2014) that allows the classification of individuals based on the strength of their preferences. The count of binary choices is compared to the random choice (equal to 6 in our experiment) using the Wilcoxon signed rank test. Our findings indicate that the subject choices in the Fork Game generally align with risk aversion (mean $= 7.90$, $p - value < 0.001$) and prudence (mean $= 9.22$, $p - value < 0.001$). We observe a modest tendency towards intemperance, with subjects making an average of 5.64 temperate choices ($p - value = 0.028$). The high prevalence of prudent choices and the varying patterns in risk aversion and temperance are consistent with the theoretical predictions of mixed risk preferences.

Figure 12 illustrates the distribution of the total number of risk-averse, prudent, and temperate choices observed in our experiment. The distribution of risk-averse choices has a peak at 7, accounting for 15.3% of the total observations. The median of this distribution is 8, indicating that a substantial majority of participants (approximately 71%) made risk-averse choices more than six times. In the case of prudent choices, the distribution shows a mode at 12, with a median of 10. About 84% of subjects made strictly more than six prudent choices, indicating a high prevalence of prudence among participants. Regarding temperance, the mode is observed at 4, and the median is at 5. Only

[5]The code for the experimental software is available at https://github.com/emrergin/prudence-labversion. A version including translations of all instructions can be found at https://github.com/emrergin/prudencetemperance.

[6]The Consumer Price Index (base: June 2003) was 207.8 in June 2012, 992.4 in June 2022, and 2353.8 in June 2024. During the first phase, 1 USD was approximately 19 TL, and a meal at the school cafeteria cost around 7.50 TL. In the second phase, 1 USD was about 34 TL, and the cafeteria meal cost approximately 30 TL.

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Table 1 Choices: risk aversion, prudence and temperance

	Mean	Standard deviation	Min	Max	Number of obs.
Risk aversion	7.90***	2.74	0	12	183
Prudence	9.22***	2.96	0	12	183
Temperance	5.64**	2.94	0	12	183

Notes: We report risk-averse, prudent, and temperate choices in our experiment. **, and *** indicate the average number of choices significantly different from random choice (6 in our experiment), at 0.05, and 0.01 respectively.(Wilcoxon test)

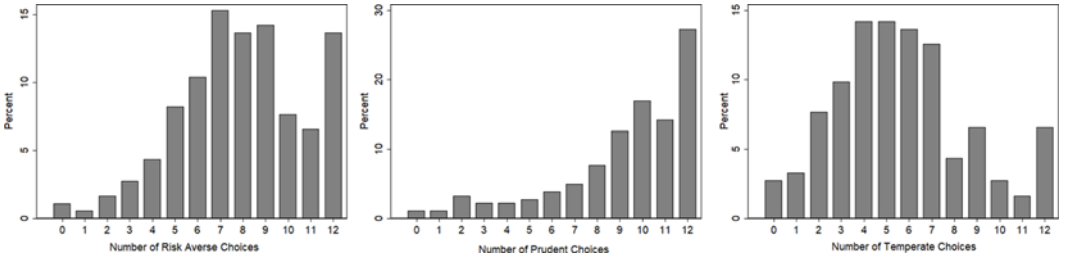


Fig. 12 Histogram of choices. (a) Risk aversion. (b) Prudence. (c) Temperance

Table 2 Rank correlations between risk aversion, prudence and temperance

	Risk aversion	Prudence
Prudence	−0.004	
Temperance	0.371***	0.083

Notes: Spearman rank correlations. *** indicates significance at 0.01 level.

35% of subjects made more than six temperate choices in this treatment, indicating lower levels of temperance compared to risk aversion and prudence.

In Table 2, the Spearman rank correlations between risk-averse, prudent, and temperate choices at the subject level are presented. We observe a positive and significant correlation between risk aversion and temperance (correlation coefficient = 0.371, $p < 0.01$). However, there is a weak and insignificant correlation between risk aversion and prudence (−0.004), as well as between prudence and temperance (0.083). Figure 13 plots the number of prudent and temperate choices based on the number of risk-averse choices. The plots show a weak and nonmonotonic relationship between risk aversion and prudence, with relatively risk-loving subjects slightly more prudent. Furthermore, the number of temperate choices tends to increase as the number of risk-averse choices increases, confirming the positive and significant correlation between risk aversion and temperance.

Table 3 presents a similar analysis, showing the average number of prudent and temperate choices based on the quantiles of the risk-averse choice distributions. The results indicate that across all four quantiles of risk distribution, the number of prudent choices ranges from 8.92 to 9.65, significantly higher than what would be expected from random choice (6 out of 12, $p < 0.01$ for all quantiles). This suggests that prudence prevails in our sample, regardless of the extent of risk aversion. On the other hand, temperance appears to be related to the degree of risk aversion: the number of temperate choices increases monotonically from 4.26 in the first quantile to 7.38 in the fourth quantile. The correlation between risk aversion and temperance is further evidenced by the significance of temperate choices in the upper quantiles ($p < 0.05$ for the fourth quantile).

We extend our analysis by employing parametric methods to explore the determinants of subject choices in our experiment. Specifically, we estimate random effects panel logit regressions, where

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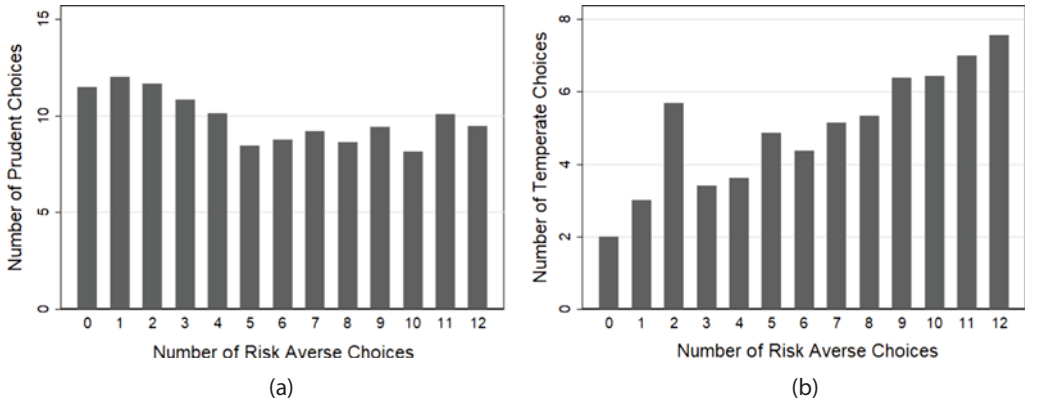


Fig. 13 Prudence-temperance by the number of risk averse choices. (a) Prudence. (b) Temperance

Table 3 Prudence and temperance by quantiles of the risk averse choices

	Risk aversion quantiles			
	First quantile	Second quantile	Third quantile	Fourth quantile
Prudence	9.40***	8.92***	8.98***	9.65***
Temperance	4.26***	5.23**	6.40	7.38**

Notes: Average number of prudent and temperant choices for each risk aversion quantile. *, **, and *** indicate the average number of choices significantly different from random choice (6 in our experiment), at 0.1, 0.05, and 0.01 respectively. (Wilcoxon test.)

the binary dependent variables indicate risk-averse, prudent, and temperate choices in each task. These choices are modeled as functions of demographic characteristics and responses to the Global Preferences Survey (GPS), as described in Falk et al. (2018, 2022). Additionally, we incorporate the endowment-to-risk ratio as a covariate to account for potential wealth effects, following the approach of Noussair et al. (2014). Background risk, defined as the absolute size of the zero-mean risk, is also included to examine its influence on prudent and temperate choices. Separate models are estimated for prudent and temperate choices to capture potential differences in how these preferences are influenced by wealth and risk considerations.

Table 4 summarizes the explanatory variables included in our regressions. The average age of participants is 21.35 years, with a standard deviation of 2.08, reflecting a relatively homogeneous sample drawn from a student population. Approximately 44% of participants identify as female. The average grade point average (GPA) is 2.88 on a 4.0 scale and around 20% of participants report prior exposure to economic experiments, captured by the variable *Experience*. Another key variable, *Econ*, represents the number of economics courses taken by participants, which is capped at 4.

In the second block of Table 4, we display information on the GPS items, which are used to elicit risk and time preferences, positive and negative reciprocity, altruism, and trust. There are 12 questions in the survey to elicit preferences of individuals in these five separate domains. More specifically, the first two variables, "Willingness to take risks" and "staircase risk," are designed to elicit risk preferences. The variables "Staircase patience" and "Willingness to give up something today" are used to assess the time preferences (i.e., the level of patience) of the participants. "Willingness to return a favor" and "Size of gift" measure positive reciprocity, while "Willingness to punish if oneself treated unfairly," "Willingness to punish if other treated unfairly," and "Willingness to take revenge" are used to construct negative reciprocity. "Willingness to give for good causes" and "Hypothetical donation" are used to elicit preferences for altruism. Finally, trust is measured based on a self-assessment question asking participants to rate their belief that people have only the best intentions, on a level between 1–10.

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Table 4 Summary statistics

	Mean	Std	Min	Max	N
Demographics					
Age	21.35	2.08	17	33	183
Female	0.44	0.50	0	1	183
GPA	2.88	0.53	1	3.5	183
Experience	0.20	0.40	0	1	183
Econ	2.16	1.50	0	4	183
Difficulty	3.70	2.13	0	8	183
Confidence	5.60	2.37	0	10	183
GPS survey					
Will. to take risks	4.98	2.16	0	10	183
Staircase risk	13.28	5.11	1	28	183
Staircase patience	10.44	10.56	1	32	183
Will. to give up sth. today	6.26	2.09	1	10	183
Will. to return favor	8.66	1.16	5	10	183
Size of gift	7.02	3.05	0	12	183
Trust	3.72	2.51	0	10	183
Will. to punish if oneself treated unfairly	6.80	2.28	0	10	183
Will. to punish if other treated unfairly	6.25	2.28	0	10	183
Will. to take revenge	5.33	2.71	0	10	183
Will. to give to good causes	6.91	2.29	0	10	183
Hypoth. donation	449	474	0	2500	183
Subjective Math Skill	6.92	2.01	0	10	183

Notes: Summary of survey responses for the participants in the Fork Game. See the Online Appendix for the details.

Following the methodology outlined in Falk et al. (2018), we standardized the responses to the survey items using the means and standard deviations from our sample. We then multiplied these standardized responses by their respective coefficients, as specified in Falk et al. (2018), and summed them to obtain the final score for each GPS item. (Further details can be found in the Online Appendix.) While we also included two additional items from the survey, namely "Subjective math skills" and the subject's self-assessment of the statement "I tend to postpone tasks even if I know it would be better to do them right away," these are not used in the construction of the main preference items. In our regressions, we included "Subjective math skills" as an explanatory variable.

We present our results in Table 5. The GPS risk variable is negatively associated with both risk-averse and temperate choices. However, we do not observe a significant effect on prudence. The results support the correlation between risk aversion and temperance, indicating that risk-averse individuals are more likely to make temperate choices, whereas risk-loving individuals are less likely to make temperate decisions. Higher GPA levels are positively associated with the likelihood of making risk-averse choices, while the perceived difficulty of the experiment has a slightly negative effect on temperance. None of the preference measures elicited through GPS show a significant association with prudence. However, GPS patience and altruism are positively correlated with the likelihood of making temperate choices. Columns 2b, 3b, and 3c of Table 5 present the results of models that include the endowment-to-risk ratio as an additional covariate for prudence and temperance. We define the risk ratio as the

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Table 5 Fork Game: Effect of preference measures, demographics and risk-to-endowment ratio on risk-averse, prudent and temperate choices

	(1) Risk	(2a) Prudent choice	(2b) Prudent choice	(3a) Temperate choice	(3b) Temperate choice	(3c) Temperate choice
Risk ratio (%)			0.001		0.003**	0.003***
			(0.001)		(0.001)	(0.001)
Background risk						0.006
						(0.007)
Female	−0.012	0.002	0.002	0.090**	0.091**	0.091**
	(0.033)	(0.041)	(0.041)	(0.038)	(0.038)	(0.038)
Age	−0.001	0.009	0.009	0.002	0.002	−0.001
	(0.009)	(0.008)	(0.008)	(0.008)	(0.001)	(0.001)
Difficulty	0.006	−0.011	−0.011	−0.006	−0.006	−0.006
	(0.008)	(0.009)	(0.009)	(0.009)	(0.010)	(0.009)
GPA	0.070**	0.002	0.002	0.042	0.042	0.042
	(0.028)	(0.033)	(0.034)	(0.039)	(0.039)	(0.039)
Experience	0.021	−0.056	−0.056	−0.064	−0.064	−0.064
	(0.038)	(0.047)	(0.046)	(0.043)	(0.044)	(0.044)
Econ	0.003	0.011	0.011	0.006	0.006	0.006
	(0.010)	(0.013)	(0.013)	(0.011)	(0.011)	(0.011)
Sure	0.015*	0.009	0.009	0.016*	0.016*	0.016*
	(0.008)	(0.008)	(0.008)	(0.009)	(0.009)	(0.009)
GPS risk	−0.134***	0.012	0.012	−0.091***	−0.091***	−0.091***
	(0.019)	(0.021)	(0.021)	(0.023)	(0.024)	(0.024)
GPS patience	−0.009	−0.015	−0.016	0.026	0.026	0.026
	(0.019)	(0.023)	(0.023)	(0.023)	(0.024)	(0.024)
GPS trust	−0.029**	0.010	0.010	0.008	0.008	0.008
	(0.014)	(0.019)	(0.019)	0.016	(0.016)	(0.016)
GPS positive reciprocity	−0.018	−0.054*	−0.054*	−0.035	−0.035	−0.035
	(0.022)	(0.029)	(0.028)	0.026	(0.026)	(0.026)
GPS negative reciprocity	0.014	−0.006	−0.006	0.030	0.030	0.012
	(0.019)	(0.027)	(0.027)	0.026	(0.026)	(0.035)
GPS altruism	0.023	0.011	0.011	0.057**	0.057**	0.057**
	(0.019)	(0.024)	(0.024)	(0.023)	(0.023)	(0.024)
Math skill	−0.002	0.009	0.009	−0.039*	−0.04*	−0.04*
	(0.019)	(0.023)	(0.023)	(0.021)	(0.021)	(0.021)
Number of subjects	183	183	183	183	183	183
Number of obs.	2196	2196	2196	2196	2196	2196

Notes: Random effects panel logit regressions for the choices in the Risk, Prudence, and Temperance treatments in the Fork Game. The risk ratio is calculated as the ratio of zero-mean risk to the expected value of the prospect and background risk is the absolute size of the zero-mean risk. Marginal effects are reported. Error terms are clustered at the participant level. *, **, and *** indicate significance at 0.1, 0.05, and 0.01 levels.

ratio of the size of the zero-mean risk to the expected value of the prospect. For the temperance task, we first calculate the background risk by fixing the zero-mean risk, which is larger (or equal) in size, and then use the other zero-mean risk to calculate the risk ratio. In column 2b, we examine the relationship between prudent choices and the risk ratio, given in percentage points. We find no significant evidence of decreasing/increasing absolute prudence in our sample. In columns 3b and 3c, we include

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the endowment-to-risk ratio and the ratio along with background risk in our regression analysis. Consistent with the findings of Noussair et al. (2014), our results indicate evidence of decreasing absolute temperance, as indicated by the positive and significant effect of the ratio on the likelihood of making temperate choices. Specifically, a one percentage point increase in the ratio leads to a 0.3 percentage point increase in the likelihood of making a temperate choice.

By incorporating survey-based measures of risk aversion, we provide additional validation of the observed link between temperance and risk aversion. This link aligns with the theoretical constructs of mixed risk aversion (Caballé and Pomansky, 1996) and mixed risk loving (Crainich et al., 2013), which describe the alternating signs of higher-order derivatives of utility functions. Mixed risk-averse individuals tend to make risk-averse, prudent, and temperate choices, while mixed risk-loving individuals are characterized by risk-loving, prudent, and intemperate choices.

An interesting theoretical implication of these constructs is that individuals with mixed risk-averse preferences tend to make not only risk-averse choices, but also prudent and temperate choices. In contrast, individuals with mixed risk-loving preferences would make risk-loving, prudent, and intemperate choices. Given that both types share similar attitudes towards prudence, we would expect prudence to be more prevalent in the population compared to temperance, and a positive association between risk aversion, temperance, and the general willingness to take risks. These are consistent with our experimental findings described above.

7. Comparison with the experimental tasks in the literature

To provide a broader comparison of methodologies, we carried out experiments that applied the designs of both Bleichrodt and Bruggen (2022) and Noussair et al. (2014), using the same lottery settings. In this section, as well as in the Online Appendix, the terms Bleichrodt and Bruggen (2022) and Noussair et al. (2014) refer to the sessions we executed using these designs and the resulting data, rather than their original datasets.

The comparison with our replication of Bleichrodt and Bruggen (2022) and our replication of Noussair et al. (2014) focuses on three aspects: the prevalence of risk-averse, prudent, and temperate choices; the distributional patterns of these choices; and operational aspects of the experiments, including task difficulty and completion times.

The results presented in Table 6 show broadly consistent patterns of risk preferences across all three experimental designs. In the Fork Game, subjects make risk-averse choices in 7.90 out of 12 decisions, comparable to 7.98 in Bleichrodt and Bruggen (2022) and 7.66 in Noussair et al. (2014). As shown in Table 7, these differences are not statistically significant ($p - value = 0.71$ and $p - value = 0.56$, respectively). A notable difference emerges in prudence choices: subjects in the Fork Game and Noussair et al. (2014) exhibit higher levels of prudence (9.22 and 9.25 respectively) compared to Bleichrodt and Bruggen (2022) (6.90). For temperance, all three designs yield similar results, with subjects making temperate choices in approximately 5-6 out of 12 decisions, with no statistically significant differences between the designs.

The distribution of choices, illustrated in Figure 14, reveals several interesting patterns across the three designs. Panel A shows that the Fork Game exhibits a right-skewed distribution for risk-averse choices, with a peak at 7 choices (15.3% of observations) and about 71% of subjects making more than six risk-averse choices. This pattern aligns with Bleichrodt and Bruggen (2022) and Noussair et al. (2014), though with slightly different concentrations around the peak. For prudent choices (Panel B), the Fork Game and Noussair et al. (2014) show similar distributions with pronounced peaks at higher values, with approximately 84% of subjects making more than six prudent choices. The distribution of temperate choices (Panel C) shows consistency across all three designs, with modes typically falling between 4–5 choices.

While all three experiments use the same underlying lottery structures derived from Eeckhoudt and Schlesinger (2006), the visual representation and framing differ. The Fork Game presents choices

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Table 6 Choices: Risk Aversion, Prudence and Temperance Across Experimental Designs

	Mean	Std	Min	Max	N
Panel A: Fork Game					
Risk aversion	7.90***	2.74	0	12	183
Prudence	9.22***	2.96	0	12	183
Temperance	5.64*	2.94	0	12	183
Panel B: Our replication of Bleichrodt and Bruggen (2022)					
Risk aversion	7.98***	2.78	0	12	112
Prudence	6.90*	3.99	0	12	112
Temperance	5.33*	3.19	0	12	112
Panel C: Our replication of Noussair et al. (2014)					
Risk Aversion	7.66***	2.95	0	12	107
Prudence	9.25***	2.96	0	12	107
Temperance	5.29*	2.9	0	12	107

Notes: We report risk-averse, prudent, and temperate choices for each experiment: Fork Game, our replication of Bleichrodt and Bruggen (2022) and our replication of Noussair et al. (2014). *, and *** indicate the average number of choices significantly different from random choice (6 in our experiment), at 0.1, and 0.01 respectively.(Wilcoxon test)

Table 7 Comparison of choices between designs

	Risk-averse choices	Prudent choices	Temperate choices
Fork Game versus our replication of Noussair et al. (2014)	7.90/7.66 ($p = 0.56$)	9.22/9.25 ($p = 0.91$)	5.64/5.29 ($p = 0.50$)
Fork Game versus. our replication of Bleichrodt and Bruggen (2022)	7.90/7.98 ($p = 0.71$)	9.22/6.90 ($p < 0.001$)	5.64/5.33 ($p = 0.44$)
Our replication of Noussair et al. (2014) versus. our replication of Bleichrodt and Bruggen (2022)	7.98/7.66 ($p = 0.42$)	9.25/6.90 ($p < 0.001$)	5.29/5.33 ($p = 0.99$)

Notes: This table reports the pairwise comparisons of risk-averse, prudent, and temperate choices between the three experimental designs, Fork Game and our replication of Bleichrodt and Bruggen (2022) and Noussair et al. (2014). The first numbers in each cell represent the mean number of choices for each design, and p-values from Wilcoxon rank-sum tests are reported in parentheses.

through the physical placement of forked pipes, Bleichrodt and Bruggen (2022) use colored balls and urns, and Noussair et al. (2014) employ a decision-tree representation. The similarity between prudence levels in the Fork Game and Noussair et al. (2014), despite their different visual approaches, suggests that certain presentation methods might facilitate the understanding of prudence concepts differently.

Our findings can be compared with those reported in Haering et al. (2020), who study the consistency of higher-order risk preferences across different presentation formats. For risk aversion, we observe similar levels, with approximately 66% risk-averse choices in our Fork Game compared to their 65–70%. However, we find somewhat stronger evidence of prudence, with about 77% prudent choices compared to their 70–75%. This pattern reverses for temperance, where we observe lower levels (47% temperate choices) compared to their findings of about 55%. Differences in prudence and temperance levels between studies may reflect the impact of experimental design on preference elicitation, a point also emphasized by Haering et al. (2020) in their comparison of compound versus reduced-form lottery presentations. In particular, our finding that the Fork Game and Noussair et al. (2014) exhibit higher levels of prudence than Bleichrodt and Bruggen (2022) aligns with their observation that the presentation format can significantly influence the elicitation of higher-order risk preferences.

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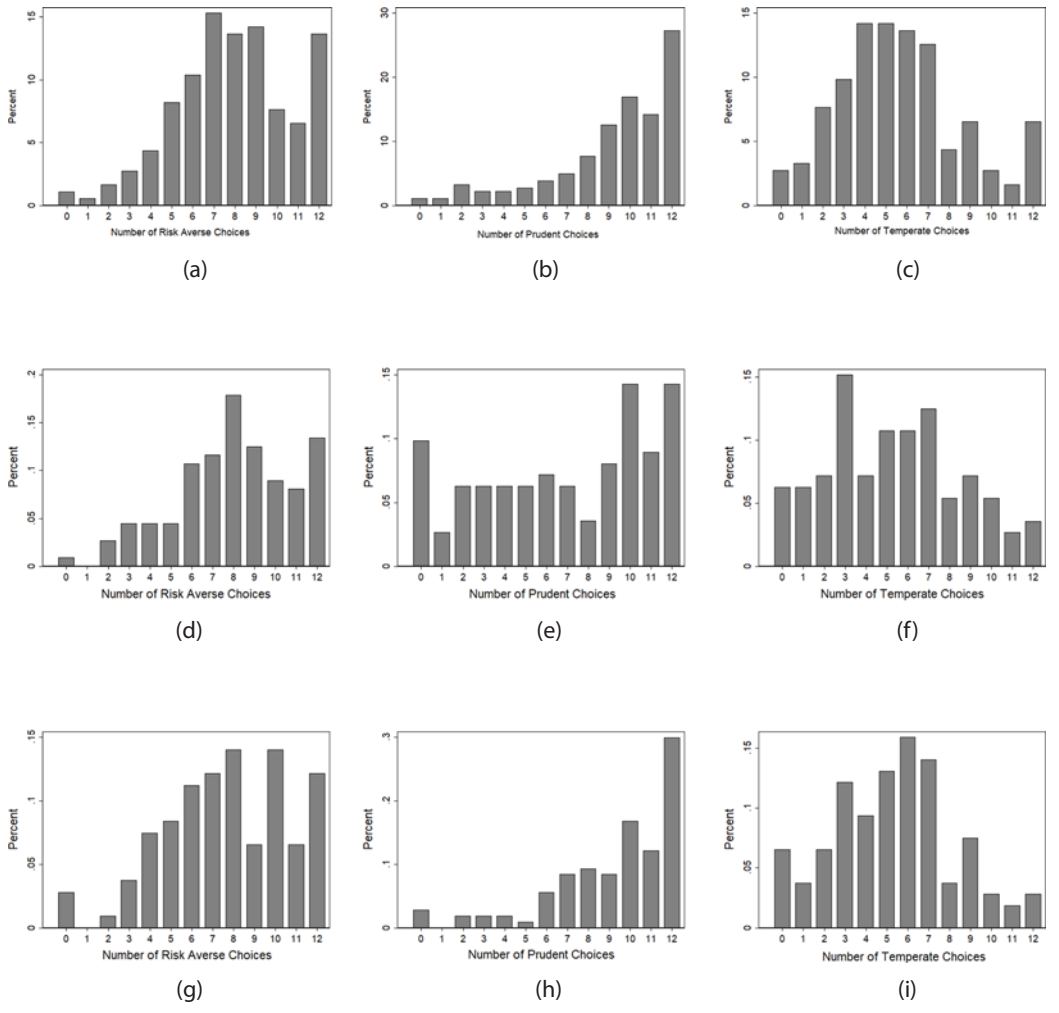


Fig. 14 Histogram of choices across different experiments. Panel A: Fork Game (a) Risk aversion. (b) Prudence. (c) Temperance. Panel B: Our replication of Bleichrodt and Bruggen (2022): (d) Risk aversion. (e) Prudence. (f) Temperance. Panel C: Our replication of Noussair et al. (2014): (g) Risk aversion. (h) Prudence. (i) Temperance.

Regarding operational aspects, as reported in Table 8, Table 9 subjects rate the perceived difficulty of the Fork Game at 3.70 (on a scale of 0–10) compared to 4.20 in Noussair et al. (2014) and 4.38 in Bleichrodt and Bruggen (2022). Pairwise comparisons using Wilcoxon tests reveal that the Fork Game's lower difficulty rating is statistically significant when compared to Bleichrodt and Bruggen (2022) ($p - value = 0.01$) and marginally significant compared to Noussair et al. (2014) ($p - value = 0.09$). The difference between Noussair et al. (2014) and Bleichrodt and Bruggen (2022) is not statistically significant ($p - value = 0.50$).

The analysis of completion times reveals varying patterns across treatments as shown in Table 9 and illustrated in Figure 15. For risk aversion tasks, average response times are relatively similar across all three designs (5.50, 5.15, and 4.68 seconds for Fork Game, Bleichrodt and Bruggen (2022), and Noussair et al. (2014) respectively). The average round time in the Fork Game (6.58 seconds) is lower than both Noussair et al. (2014) (6.89 seconds) and Bleichrodt and Bruggen (2022) (7.96 seconds). While the difference between the Fork Game and Noussair et al. (2014) is not statistically significant

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Table 8 Operational aspects of designs

	Mean	Std	Min	Max	N
Panel A: Self-reported difficulty					
Fork Game	3.70	2.13	0	8	183
Our replication of Noussair et al. (2014)	4.20	2.21	0	10	107
Our replication of Bleichrodt and Bruggen (2022)	4.38	2.01	0	9	112
Panel B: Average round time (seconds)					
Fork Game	6.58	2.81	3.06	18.11	183
Our replication of Noussair et al. (2014)	6.89	4.34	2.3	25.97	107
Our replication of Bleichrodt and Bruggen (2022)	7.96	3.61	3.09	25.20	112

Notes: This table reports operational measures for each experimental design, Fork Game and our replication of Bleichrodt and Bruggen (2022) and Noussair et al. (2014). Panel A shows the self-reported difficulty ratings on a scale of 0–10, where higher numbers indicate greater perceived difficulty. Panel B shows the average time taken to complete a round in seconds, including the entire decision-making process and outcome resolution.

Table 9 Average round times

	Mean	StD	Min	Max	N
Average round time (in seconds)					
Fork Game					
Risk	5.50	2.26	2.77	16.92	183
Prudence	5.65	3.75	1.94	30.33	183
Temperance	8.60	4.37	3.6	30.54	183
Our replication of Noussair et al. (2014)					
Risk	4.68	2.27	2.16	15.65	107
Prudence	6.86	5.69	2.15	49.55	107
Temperance	9.12	7.26	1.6	47.33	107
Our replication of Bleichrodt and Bruggen (2022)					
Risk	5.15	1.84	2.67	11.7	112
Prudence	8.34	4.86	2.26	30.28	112
Temperance	10.38	5.78	2.4	27.27	112

Notes: This table reports detailed round completion times for each treatment across experimental designs, Fork Game and our replication of Bleichrodt and Bruggen (2022) and Noussair et al. (2014). All times are reported in seconds and include both the decision-making process and outcome resolution. The summary statistics include all rounds for all participants in each treatment. For task-specific comparisons, see Tables 8–10 in the Online Appendix.

($p - value = 0.32$), both designs show significantly faster completion times compared to Bleichrodt and Bruggen (2022) ($p - value < 0.001$ for both comparisons).

To examine these differences more rigorously, pairwise comparisons of completion times for each task (1–12 in Bleichrodt and Bruggen (2022)) were performed between designs.[7] The results indicate that differences in completion times are more pronounced in prudence treatment, where they are statistically significant for most tasks ($p - value < 0.01$). The differences are less consistent in the temperance treatment, though the Fork Game generally maintains faster completion times.

Further evidence on the robustness of our findings is provided in the Online Appendix, where we present detailed task-by-task comparisons and regression analyses. The parameter-matched comparisons reveal consistent patterns in elicited preferences across different presentation formats.

[7]Detailed results of these pairwise comparisons for risk aversion, prudence, and temperance tasks are reported in Tables 8, 9, and 10 in the Online Appendix.

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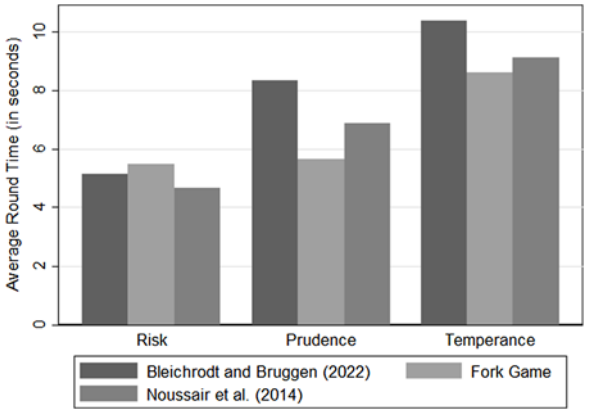


Fig. 15 Average round time for each treatment and game

Task-level analysis shows that the higher prudence levels in the Fork Game and our replication of Noussair et al. (2014) persist across different parameter combinations, suggesting this finding reflects a genuine difference in how prudence is understood across different presentation formats rather than an artifact of specific parameter choices. Regression analyses demonstrate consistent relationships between individual characteristics and risk preferences across all three experimental designs, particularly in the relationship between general risk attitudes and both risk-averse and temperate choices.

These comparisons suggest that the Fork Game maintains the ability to elicit higher-order risk preferences consistent with existing experimental designs. The strong correlation between risk-averse choices and similar patterns in temperance across all three designs provides evidence for the robustness of these elicitation methods. Differences in prudence levels and completion times between designs warrant further investigation to better understand how various presentation methods might affect the elicitation of higher-order risk preferences.

The consistency of the results in risk aversion and temperance between designs suggests that the divergence in prudence levels is not due to systematic differences in the subject groups or general risk attitudes. Rather, these differences might be attributed to specific aspects of how prudence choices are presented in each design. Further research could explore which elements of different experimental designs most effectively facilitate the elicitation of higher-order risk preferences.

8. Summary and discussion of results

In this study, we introduce the "Fork Game," a novel graphical device designed to elicit higher-order risk preferences. Our method utilizes forking pipes to represent the randomization aspect of the experiments where the resulting outcome is represented by a falling ball within those pipes, taking a randomly chosen path at the forks. In the treatments, participants are tasked with placing the pipes to construct the desired lottery, which is equivalent to the lottery pairs used in the previous literature. By visualizing the randomization aspect and the chosen lottery clearly, and designing all games as direct variations of the same underlying concepts, we can extend the games in a straightforward manner to analyze even more complex risk attitudes without significantly increasing the procedural complexity.

Tasks aimed at eliciting higher-order risk preferences often involve complex gambles, which can lead to distinct decision-making patterns compared to simplified versions of the tasks. The comprehension of the overall consequences of complex gambles poses a challenge for subjects. Additionally, task difficulty is a relevant factor to consider. In our study, we believe that our task is relatively simple,

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as the average reported difficulty is 3.7 out of 10 and no participant has reported a rating above 8 in our experiments.

Our laboratory experiments reveal consistent results with previous findings regarding higher-order risk preferences. Subjects in the "Fork Game" exhibit a high prevalence of risk aversion and prudence, while temperance is less common. Intemperate choices are observed among a significant fraction of subjects. Our results align qualitatively and quantitatively with those of Bleichrodt and Bruggen (2022), indicating a positive and statistically significant correlation between our experiment and their study. Although we do not find a significant correlation for temperance, our experiment's overall level of temperate choices is similar to that observed by Bleichrodt and Bruggen (2022).

Furthermore, we relate binary choices in risk aversion, prudence, and temperance treatments to the demographic characteristics and responses to the Global Preference Survey (GPS). We find that the GPS risk variable is negatively associated with risk-averse and temperate choice, however, we do not observe a significant effect on prudence which further validates the observed relation between risk aversion and temperance. This observation can be attributed to the presence of mixed risk aversion and mixed risk-loving preferences, which are commonly found in the population. Mixed risk-averse individuals tend to be prudent and temperate and mixed risk-loving preferences are associated with prudence and intemperance. Despite their different risk preferences, both groups show similar attitudes towards prudence, leading to a prevalence of prudence in the population, and a positive association between risk aversion, temperance, and the overall willingness to take risks. These findings are consistent with our experimental results.

Supplementary material. The supplementary material for this article can be found at https://doi.org/10.1017/eec.2025.10022.

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Competing interests. The authors declare that they have no competing interests.

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